

§1 Open Expansion, Fracture & R_t

The dynamical foundation of the Reconciliation Causal Framework, rewritten for v2.1. Activates the two co-primitives via Open Expansion, defines the energy mismatch, formalizes the fracture of H_{kin} and the $\hat{F} = F_{\text{intra}} + F_{\text{cross}}$

6 GAPS FLAGGED

decomposition, defines burden and the Tier 1/Tier 2 split, constructs the Reconciliation Propagator R_t , and derives $\gamma = \sqrt{\lambda_{\min}(\hat{F})}$ from the Lieb-Robinson bound. Incorporates the adjacency structure on $\{\hat{C}_\alpha\}$ from Jordan co-occurrence. Six inline gap flags surface the T-1/T-2 prerequisites at their point of origin.

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SUBSECTIONSC
LAYER15+
DEFINITIONS5
THEOREMS

VERSION

AUDIENCE

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v2.1 Draft (Canonical Form)

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RCF Section 1 — Open Expansion, Fracture, and the Reconciliation Propagator

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

Section 0 established the algebraic foundation (A, admissibility, GNS) and named the two structural co-primitives — causality (Lie algebra A_L) and locality (Jordan algebra A_J). These primitives are operationally inert until activated by the Open Expansion Principle, which is the subject of this section.

The v2.0 merge placed the Reconciliation Propagator R_t (§0.4 in v1.0), the Convergence Theorem (§0.5), the thin/full A_{phy} split (§0.6–0.7), the reduced algebra and Dirac bracket (§0.7b), and the Reconciliation Principle (§0.8) all within Section 0 — before causality and locality were named, and before the fracture machinery existed. This created structural problems: M_{red} was referenced but never defined (it requires the fracture and the Tier 1/Tier 2 split), Π_{net} forward-referenced K_ω from §2, and the Reconciliation Principle was stated before burden's physical interpretation was available.

What changed from v2.0

v2.1 retains the v2.0 rebirth structure (this section assembles all the displaced v1.0 §0 content in its correct emergence order) with the following corrections and additions:

1. **Open Expansion Principle reframed as Lie–Jordan activation** (§1.1). v2.0 stated the principle as a combinatorial openness postulate; v2.1 reframes it as a dynamical principle: local Jordan-built sectors continuously expand their algebraic scope to reconcile with the single global causal Lie layer. This grounds the principle in the two-primitive picture from §0.
2. **Adjacency structure on $\{\hat{C}_\alpha\}$ from Jordan co-occurrence** (§1.1.4, NEW). v2.0 had no notion of "local" for the constraint index set. v2.1 defines: two constraints $\hat{C}_\alpha, \hat{C}_\beta$ are adjacent iff $\hat{C}_\alpha \circ \hat{C}_\beta \neq 0$ (Jordan co-occurrence). This adjacency graph is the shared prerequisite for T-1 (Lieb-Robinson bounds) and T-2 (detectability lemma).
3. **$\gamma = \sqrt{\lambda_{\min}(F)}$ derivation** (§1.1.5, NEW). v2.0's Theorem 1.1.2 (finite propagation rate $c = \gamma \cdot \ell_0$) was labeled "Conditional" but the proof sketch was a dimensional argument. v2.1 proposes $\gamma := \sqrt{\lambda_{\min}(F)}$ where $\lambda_{\min}$ is the smallest non-zero eigenvalue of F . This makes $c = \ell_0 \cdot \sqrt{\lambda_{\min}(F)}$ a derived constant, and unifies T-1 and T-2 (both reduce to the spectral gap of F).
4. **F decomposition formalized** (§1.2.4, NEW). v2.0 named the decomposition $F = F_{intra} + F_{cross}$ in Remark 0.3.7 but deferred formalization. v2.1 defines it formally: $F_{intra} = \sum_i P_i F P_i$ (intra-sector fracture), $F_{cross} = \sum_{i \neq j} P_i F P_j$ (cross-sector fracture). This is the basis for the relational-burden identification in §4.4.
5. **M_{red} definition retained** (§1.4.3). v2.0's Definition 1.4.3 (M_{red} as Tier 2 sub-collective, "separation not reduction") is retained verbatim. The downstream fix to Theorem 2.7.1 (sectorwise zero-decomposition) depends on it.

6. **Convergence Theorem with detectability lemma approach** (§1.6, EXPANDED). v2.0 stated the theorem with a LaSalle invariance proof sketch (verified only in a finite-dim toy model). v2.1 expands the proof strategy: F is a frustration-free Hamiltonian; the detectability lemma (Aharonov et al. 2009) lifts local gaps to a global gap. The uniformity condition (local gap doesn't shrink under expansion) is flagged as the key remaining step.

7. **Terminology: SOE → Open Expansion, MOE → Open Extension** (retained from v2.0). The Definition 1.5.2/1.5.3 headers carry the parenthetical "(was SOE)" / "(was MOE)" for traceability.

8. **Reconciliation Principle REMOVED from §1 scope** (retained from v2.0). RP is deferred to §2.1, after burden (§1.3) and events (§1.8) are defined. The v1.0 fault — "RP was stated before burden's physical interpretation was available" — is corrected.

What this section does NOT contain (deferred)

- The Reconciliation Principle (→ §2.1)
- K_ω , the Fubini-Study metric, emergent geometry (→ §3)
- Burden-clock suppression $\alpha(B)$, duration, the "now" (→ §4)
- The burden tensor, Einstein closure, gravity (→ §4)
- Black holes, cosmology (→ §5, §6)

The test for this section: does it contain anything that depends on K_ω , the Fubini-Study metric, duration, gravity, or the Reconciliation Principle? If yes, the reordering has failed.

§1.0 Purpose: The Dynamical Foundation

Section 0 produced the algebraic substrate: A , admissibility, GNS, the algebraic F , and the two co-primitives (causality/Lie, locality/Jordan). These objects are static — they describe what the framework IS, not what it DOES. Section 1 produces the dynamical foundation: how the algebraic substrate becomes physical.

The key sequence is:

1. **Open Expansion activates the two primitives** (§1.1) — local Jordan-built sectors expand to reconcile with the global causal Lie layer.
2. **Finite expansion rate creates Lie–Jordan mismatch** (§1.1.3) — because reconciliation propagates at finite speed $c = \gamma \cdot \ell_0$, local sectors cannot adapt instantaneously.
3. **The mismatch fractures H_{kin} into sub-Hilbert spaces** (§1.2) — the global algebra cannot maintain a single unified GNS representation.
4. **F gets its physical interpretation as fracture measure** (§1.2.4) — and is decomposed into F_{intra} + F_{cross} .
5. **Burden becomes the cost of the fracture** (§1.3) — $B_\Delta = \text{Tr}(\rho \cdot F)$, now physically interpreted.

6. **Tier 1/Tier 2 split becomes visible** (§1.4) — the fracture exposes which constraints are second-class (quotiented away) and which are first-class (survive on A_{red}).
7. **A_{red} , M_{red} , Dirac bracket are defined** (§1.4.2–1.4.6) — the reduced algebra and its master constraint.
8. **R_t is constructed** (§1.5) — the three-mechanism propagator ($\text{Open Expansion} \circ \text{Open Extension} \circ \text{Dephasing}$).
9. **The Convergence Theorem makes Master-Zero a derived result** (§1.6) — R_t drives every state to $\ker(M) = \ker(F)$.
10. **The thin/full A_{phy} split is available** (§1.7) — the physical sub-algebra.
11. **Zero-preserving events and the operational causal order $\prec$ arrive** (§1.8) — the last objects needed before §2.

Every object is defined before use; the chain is acyclic. The closing Acyclicity Test (§1.9) verifies this.

§1.1 The Open Expansion Principle and the Adjacency Structure

Postulate 1.1.1 (Open Expansion Principle)

Open Expansion is the dynamical principle that activates the two structural co-primitives (causality/Lie and locality/Jordan, from §0.4):

> Local sectors (Jordan-built structures within H_{kin}) continuously expand their algebraic scope to reconcile with the single global causal layer (Lie structure).

Operationally: the algebra A is not static. Its represented sectors grow — new relations become admissible, new zero-preserving extensions open — driven by the requirement that locality (Jordan) and causality (Lie) remain jointly compatible under the Master-Zero condition $\omega(M) = 0$.

Remark 1.1.2 (Reframing from v2.0). v2.0 stated the Open Expansion Principle as a combinatorial openness postulate (" $\text{Ext}(s) \neq \emptyset$ for all physical s "). v2.1 reframes it as a dynamical principle grounded in the Lie–Jordan decomposition: the activation of the two primitives IS the foundational dynamic. The mathematics is unchanged (the propagator R_t has the same form); the physical interpretation is sharpened. The two-primitive picture from §0.4 is now the load-bearing foundation.

Definition 1.1.3 (Lie–Jordan Mismatch)

The **Lie–Jordan mismatch** at a kinematic state ω is the quantity:

$$\delta J[\omega] := J_{\text{target}} - J_{\text{current}}[\omega]$$

where $J_{\text{current}}[\omega]$ measures the current degree of Lie–Jordan compatibility at ω , and J_{target} is the target compatibility (full reconciliation). The mismatch is non-zero whenever the local Jordan sectors have not yet caught up with the global causal Lie layer's current state.

Operationally, δJ is mediated by the fracture operator F (§1.2.4): $B_{\Delta}[\rho] = \text{Tr}(\rho \cdot F)$ is the quantitative measure of the mismatch. The mismatch drives the dynamics — Open Extension (§1.5.3) is gradient

descent on B_Δ .

[GAP: v2.1 — the formal decomposition of δJ into J_{target} and J_{current} is not yet rigorous. J_{current} can be defined via B_Δ (the current burden), but $J_{\text{target}} = 0$ (full reconciliation) is only meaningful after the Convergence Theorem (§1.6) establishes the fixed point. The mismatch δJ is therefore best understood as B_Δ itself, with the "target" being $B_\Delta = 0$. This is consistent but makes $\delta J = -B_\Delta$, which is a notational refinement, not a structural issue. Flagged for cleanup in Phase C.]

Theorem 1.1.4 (Finite Propagation Rate)

[v2.1: this theorem is STRENGTHENED relative to v2.0. v2.0 labeled it "Conditional" with T-1 as the blocking dependency, but the proof sketch was a dimensional argument. v2.1 provides the derivation $\gamma = \sqrt{\lambda_{\min}(F)}$, making c a derived constant — conditional only on the spectral gap existing (which is T-2).]

The Open Expansion rate γ is derived from the spectral structure of the fracture operator:

$$\gamma := \sqrt{\lambda_{\min}(\hat{F})}$$

where $\lambda_{\min}(F)$ is the smallest non-zero eigenvalue of F (the spectral gap on the orthogonal complement of $\ker(F)$). The finite propagation speed is:

$$c = \gamma \cdot \ell_0 = \ell_0 \cdot \sqrt{\lambda_{\min}(\hat{F})}$$

This is a derived constant of the algebra, not a postulated parameter.

Proof sketch. The Open Expansion generator is $H_{\text{eff}} = \sqrt{F + \delta F}$. By the Lieb-Robinson theorem (1972), for a system with local interactions, the commutator $[A(t), B]$ stays exponentially small until t exceeds $d(A, B)/v_{\text{LR}}$, where v_{LR} is the Lieb-Robinson velocity:

$$\|[A(t), B]\| \leq C \|A\| \|B\| e^{-\mu(d(A, B) - v_{\text{LR}} t)}$$

The Lieb-Robinson velocity for $H_{\text{eff}} = \sqrt{F}$ is:

$$v_{\text{LR}} \sim \sqrt{\lambda_{\min}(\hat{F})} \sim \sqrt{\sum_{\alpha, \beta} \|\Delta_{\alpha\beta}\|}$$

On the physical sector, $F \rightarrow 0$. Near the physical sector, the relevant scale is the lowest non-zero eigenvalue $\lambda_{\min}(F)$. Therefore:

$$\gamma = \sqrt{\lambda_{\min}(\hat{F})}, \quad c = \ell_0 \cdot \gamma$$

The Lieb-Robinson bound ensures that information cannot propagate faster than $c = \gamma \cdot \ell_0$.

Remark 1.1.5 (Unification of T-1 and T-2). Theorem 1.1.4 unifies T-1 (γ derivation) and T-2 (spectral gap): both reduce to the spectral gap of F . The same gap that determines the Convergence Theorem's convergence rate (T-2, §1.6) also determines the causal speed limit (T-1, this theorem). This is a significant simplification — one mathematical object, two physical consequences.

[GAP: v2.1 — Theorem 1.1.4 depends on the Lieb-Robinson bound, which requires a notion of "local" for the constraint interactions. The adjacency structure on $\{\hat{C}_\alpha\}$ (Definition 1.1.6 below) provides this, but the Lieb-Robinson theorem's standard form assumes the interactions are defined on a lattice or graph with bounded degree. Whether RCF's adjacency graph (from Jordan co-occurrence) has bounded degree is NOT yet proven. If the degree is unbounded, the Lieb-Robinson velocity may not be well-defined, and Theorem 1.1.4 reverts to "Conditional (T-1)." See GAP-1.1.7.]

Definition 1.1.6 (Adjacency Structure on the Constraint Index Set)

[NEW in v2.1 — this is the shared prerequisite for T-1 and T-2 that was missing in v2.0.]

Two constraints $\hat{C}_\alpha, \hat{C}_\beta \in \{\hat{C}_\alpha\}_{\alpha \in I}$ are **adjacent** iff their Jordan product is non-zero:

$$\alpha \sim \beta \quad \Longleftrightarrow \quad \hat{C}_\alpha \circ \hat{C}_\beta \neq 0$$

The **constraint adjacency graph** G_I is the graph with vertex set I and edge set $\{(\alpha, \beta) : \alpha \sim \beta\}$. The **degree** Δ_α of a constraint $\hat{C}_\alpha$ is the number of constraints adjacent to it: $\Delta_\alpha = |\{\beta \in I : \alpha \sim \beta\}|$.

Remark 1.1.7 (Why Jordan co-occurrence defines adjacency). The two-primitive picture (§0.4) says locality = Jordan (the "where"-capacity). Two constraints are "in the same locality sector" — i.e., they operate on the same region of the emergent geometry — iff their Jordan product is non-zero. This is the algebraic signature of spatial adjacency. The Lie bracket $[\hat{C}_\alpha, \hat{C}_\beta]$ (which defines the compatibility bracket $\Delta_{\alpha\beta}$ and appears in F) measures obstruction, not adjacency — two constraints can have zero Lie bracket (commute) but still be adjacent (co-occur).

The adjacency graph G_I is therefore the locality structure of the constraint set, derived from the Jordan algebra. The Lie brackets are the dynamics on this graph. This is exactly the two-primitive picture: Jordan = the graph, Lie = the dynamics.

**[GAP: v2.1 — whether the adjacency graph G_I has bounded degree is NOT yet proven. Bounded degree is required for:

- The Lieb-Robinson bound (Theorem 1.1.4) — needs bounded degree to define v_{LR} .
- The detectability lemma (Theorem 1.6.1) — needs bounded degree Δ to bound the global gap as $\gamma_{\text{global}} \geq \gamma_{\text{local}}/(C \cdot \Delta)$.

The plausibility argument: if $D=3$ is fixed (§3.6), each reconciliation cell has at most 6 neighbors (the 3D kissing number), so $\Delta \leq 6$. But this argument is CIRCULAR — it uses $D=3$, which itself depends on the Jordan algebra representation theory (GAP-0.1.5). If $D=3$ is not yet proven, the bounded-degree claim is also open.

Resolution path: prove bounded degree DIRECTLY from the Jordan algebra structure, without assuming $D=3$. This may be easier than proving $D=3$ itself — it requires only that the Jordan co-occurrence graph is locally finite, not that the emergent geometry is 3-dimensional. Flagged as GAP-1.1.7, severity FATAL (blocks T-1 and T-2).]**

§1.2 The Fracture of H_{kin} and the F Decomposition

Definition 1.2.1 (Fracture of the Kinematic Hilbert Space)

The **fracture** of H_{kin} is the decomposition:

$$\mathcal{H}_{\text{kin}} \rightarrow \bigoplus_k \mathcal{H}_k \oplus \mathcal{H}_{\text{cross}}$$

where each H_k is a **sub-sector** — a maximal subspace on which the Jordan structure has locally converged toward Lie-Jordan compatibility — and H_{cross} is the residual cross-sector coherence (the off-diagonal structure connecting the sub-sectors).

Each H_k is a sector where the local Jordan algebra has achieved internal consistency: the constraints $\{\hat{C}_\alpha\}$ restricted to H_k have a well-defined compatibility structure. The fracture occurs because the finite

propagation speed $c = \gamma \cdot \ell_0$ (Theorem 1.1.4) prevents global synchronization — different regions of the algebra reach local consistency at different times, creating distinct sectors.

Remark 1.2.2 (Inversion of the v1.0 logical order). In v1.0, the fracture was a downstream consequence of the Reconciliation Principle (Corollary 0.8.3). In v2.0/v2.1, the fracture is upstream — it is what gives F its physical interpretation (§1.2.4), what creates the burden (§1.3), and what makes the Tier 1/Tier 2 split visible (§1.4). The Reconciliation Principle itself is deferred to §2. This inversion is the single largest architectural change in v2.0/v2.1 §1.

Definition 1.2.3 (Sector Projectors)

The **sector projectors** $\{P_k\}$ are the orthogonal projectors onto the sub-sectors $\{H_k\}$:

$$P_k: \mathcal{H}_{\{k\}} \rightarrow \mathcal{H}_k, \quad \sum_k P_k + P_{\text{cross}} = \mathbf{1}$$

where P_{cross} projects onto H_{cross} . The projectors are orthogonal ($P_k P_j = \delta_{\{kj\}} P_k$ for k, j in the sector index) and complete.

Theorem 1.2.4 (F Physical Interpretation and Decomposition)

[v2.1: this theorem is NEW. v2.0 named the F decomposition in Remark 0.3.7 but deferred formalization. v2.1 formalizes it here, as the basis for the relational-burden identification in §4.4.]

The fracture operator F (Definition 0.3.5) is physically interpreted as the **fracture measure** — the quantitative measure of the Lie–Jordan mismatch that the fracture resolves. The quantity $B_\Delta[\rho] = \text{Tr}(\rho \cdot F)$, previously an algebraic non-negative number (Remark 0.3.6), is now named **burden** (defined formally in §1.3).

F decomposes via the sector projectors:

$$\hat{F} = \hat{F}_{\text{intra}} + \hat{F}_{\text{cross}}$$

where:

- **Intra-sector fracture:** $F_{\text{intra}} := \sum_k P_k F P_k$ — captures constraint violations WITHIN each sub-sector.
- **Cross-sector fracture:** $F_{\text{cross}} := \sum_{k \neq j} P_k F P_j$ — captures the cost of maintaining coherence ACROSS sectors.

Both terms are positive: $F_{\text{intra}} \geq 0$ and $F_{\text{cross}} \geq 0$ (since $F \geq 0$ and the projectors are orthogonal).

Proof. The decomposition follows from inserting $1 = \sum_k P_k + P_{\text{cross}}$ on both sides of F :

$$\hat{F} = \left(\sum_k P_k + P_{\text{cross}} \right) \hat{F} \left(\sum_j P_j + P_{\text{cross}} \right) = \sum_k P_k F P_k + \sum_{k \neq j} P_k F P_j + P_{\text{cross}} F P_{\text{cross}}$$

The diagonal terms ($k = j$) give $F_{\text{intra}} = \sum_k P_k F P_k$. The off-diagonal terms ($k \neq j$) give $F_{\text{cross}} = \sum_{k \neq j} P_k F P_j$. The P_{cross} terms are absorbed into the appropriate sector or treated as residual. Positivity follows from $F \geq 0$ and the fact that $P_k F P_k \geq 0$ (as a compression of a positive operator).

Remark 1.2.5 (Physical interpretation of the decomposition). The decomposition $F = F_{\text{intra}} + F_{\text{cross}}$ has a direct physical meaning:

- **F_intra** is the cost of maintaining zero-preservation within a single localized sub-algebra A_k . It is the mode/existence burden — what sources baryonic matter ($m^2 \equiv B_0$, §2.5.1).

- **F_cross** is the cost of maintaining coherence between sectors k and j via the cross-transition operator P_{kj} . It is the relational burden — what sources dark matter (§4.4, §5).

This decomposition is the basis for the v2.1 identification of relational burden as cross-sector linking cost. The dark-matter mechanism (§4.4.3) becomes: dark matter = $\text{Tr}(\rho \cdot F_{\text{cross}})$, the cross-sector fracture cost. See §4.4 for the full development.

Remark 1.2.6 (Connection to the cross-extension network operator). The cross-sector fracture F_{cross} is related to the cross-extension network operator Π_{net} (defined in §3.9, after K_ω is available). The relational burden channel $\Theta^{\text{rel}} = [\hat{C}_\alpha, \Pi_{\text{net}}]$ (§4.4.1) is the network-weighted sum of cross-sector commutators:

$$[\hat{C}_\alpha, \hat{\Pi}_{\text{net}}] = \sum_{k \neq j} K_\omega(k, j) \cdot [\hat{C}_\alpha, P_{kj}]$$

This connects F_{cross} (the operator decomposition) to Π_{net} (the network structure). The identification $B_{\text{rel}} = \text{Tr}(\rho \cdot F_{\text{cross}}) = \sum_{k \neq j} K_\omega(k, j) \cdot B_{\text{cross}}(k, j)$ is formalized in §4.4.

[GAP: v2.1 — the F decomposition requires the sector projectors $\{P_k\}$, which require the fracture (Definition 1.2.1). The fracture is defined here but its EXISTENCE (whether H_{kin} actually decomposes into well-defined sub-sectors) depends on the Convergence Theorem (§1.6), which depends on T-2. If T-2 fails, the fracture may not be well-defined, and the F decomposition stalls. This is the same T-2 dependency flagged in GAP-0.3.7. Flagged as GAP-1.2.4, severity MATERIAL (the decomposition is algebraically sound; the issue is whether the sectors exist physically).]

§1.3 Burden and Burden Linearity

Definition 1.3.1 (Burden)

Burden is the physical quantity:

$$B_{\Delta}[\rho] := \text{Tr}(\rho \cdot \hat{F})$$

where ρ is a density operator on H_{kin} and F is the fracture operator (Definition 0.3.5). Burden is the cost of the Lie–Jordan mismatch — the measure of reconciliation resistance. It is non-negative ($B_{\Delta} \geq 0$ since $F \geq 0$ and $\rho \geq 0$) and vanishes on the physical sector ($B_{\Delta} = 0$ iff $\rho \in \ker(F)$, i.e., iff $\omega(M) = 0$).

Remark 1.3.2 (Burden as fracture cost). The narrative sharpening from v2.0: burden is not just "an algebraic quantity"; it is the COST OF THE FRACTURE. The fracture (§1.2) decomposes H_{kin} into sub-sectors because the finite propagation speed prevents global reconciliation. Burden measures how much reconciliation resistance remains — how far the current state is from the fully reconciled (Master-Zero) state. This physical interpretation is what makes the Reconciliation Principle (§2.1) well-posed: RP minimizes burden, which is now a meaningful physical quantity.

Property 1.3.3 (Burden Linearity)

Burden is LINEAR in the density operator:

$$B_{\Delta}[\sum_k p_k \rho_k] = \sum_k p_k \cdot B_{\Delta}[\rho_k]$$

Proof. $B_\Delta[\sum_k p_k \rho_k] = \text{Tr}((\sum_k p_k \rho_k) \cdot F) = \sum_k p_k \cdot \text{Tr}(\rho_k \cdot F) = \sum_k p_k \cdot B_\Delta[\rho_k]$. The second equality uses the linearity of the trace.

Remark 1.3.4 (Burden linearity is algebraic, not probabilistic). The identity $B_\Delta[\sum_k p_k \rho_k] = \sum_k p_k \cdot B_\Delta[\rho_k]$ holds because Tr is linear and F is fixed — it is an ALGEBRAIC IDENTITY, not a probabilistic averaging. This distinction is the load-bearing property that licenses the FIREWALL guardrail (§2.8.1): branch weights p_k are probabilistic (Layer A), but burden B_Δ is algebraic (Layers A/B/C/Q'). The burden-linearity identity does NOT conflate the two — it is a statement about the linear functional $\text{Tr}(\cdot F)$, not about probabilistic averaging over outcomes.

Remark 1.3.5 (Decomposition: $B_{\text{intra}} + B_{\text{rel}}$). Via the F decomposition (Theorem 1.2.4):

$$B_\Delta[\rho] = B_{\text{intra}}[\rho] + B_{\text{rel}}[\rho]$$

where $B_{\text{intra}}[\rho] = \text{Tr}(\rho \cdot F_{\text{intra}})$ (mode/existence burden) and $B_{\text{rel}}[\rho] = \text{Tr}(\rho \cdot F_{\text{cross}})$ (relational burden). Both terms inherit burden linearity (Property 1.3.3). This decomposition is the algebraic foundation for the three-tier burden decomposition in §4.4 (mode + relational + interaction).

§1.4 The Tier 1/Tier 2 Split and the Reduced Algebra

Definition 1.4.1 (Tier 1 / Tier 2 Separation)

The fracture (§1.2) makes the Tier 1/Tier 2 separation of the constraint set operationally meaningful:

- **Tier 1 (second-class constraints):** $\{K_a\}_{a \in I_1}$ — constraints whose compatibility matrix $\Delta^{(1)}_{ab} = [K_a, K_b]$ is invertible. These constraints are quotiented away by the Dirac bracket (§1.4.5).

- **Tier 2 (first-class constraints):** $\{D_i\}_{i \in I_2}$ — constraints whose compatibility matrix $\Delta^{(2)}_{ij} = [D_i, D_j]$ vanishes (or is degenerate). These constraints survive on the reduced algebra A_{red} .

The partition $\{\hat{C}_\alpha\} = \{K_a\} \sqcup \{D_i\}$ is determined by the rank structure of the compatibility matrix $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$, which is made visible by the fracture.

Definition 1.4.2 (Reduced Algebra)

The **Tier 1 ideal** is the two-sided $*$ -ideal generated by the Tier 1 constraints:

$$\mathcal{I}_K := \langle \hat{K}_a : a \in I_1 \rangle$$

The **reduced algebra** is the quotient:

$$A_{\text{red}} := A / \mathcal{I}_K$$

Elements of A_{red} are equivalence classes $[A] = A + \mathcal{I}_K$. The reduced algebra inherits the *-algebra structure from A (the quotient is well-defined because $\mathcal{I}_K$ is a a -ideal).*

Definition 1.4.3 (M_{red} — Tier 2 Sub-Collective)

[Retained verbatim from v2.0. This is the definition that closes the v1.0 " M_{red} gap."]

The **reduced master constraint** is the Tier 2 sub-collective of M :

$$\hat{M}_{\text{red}} := \sum_{i \in I_2} w_i \cdot \hat{D}_i^{\dagger} \hat{D}_i \in A_{\text{red}}$$

M_{red} is NOT a "reduction" of M ; it is the SEPARATION of M into its Tier 1 and Tier 2 sub-collectives. The full $M = M_{\text{Tier1}} + M_{\text{red}}$; on A_{red} , only M_{red} survives (the Tier 1 part is quotiented away by the Dirac bracket, §1.4.5).

Remark 1.4.4 (Why "separation, not reduction"). The framing "separation, not reduction" pre-empts a reader misreading of M_{red} as a "reduced" version of M . M_{red} is not an approximation or a truncation — it is the Tier 2 component of the same collective, isolated by the fracture. The full M is still $M_{\text{Tier1}} + M_{\text{red}}$; M_{red} is what remains on A_{red} after the Tier 1 ideal is quotiented away.

Remark 1.4.5 (Connection to field admissibility). The field admissibility condition in §2.3.1 (Definition 2.3.1) uses M_{red} on A_{red} , not the full M on A :

$$\hat{\phi}(f) \cdot \ker(\hat{M}_{\text{red}}) \subseteq \ker(\hat{M}_{\text{red}})$$

This resolves the v1.0 gap where M_{red} was referenced (in §4.1.2's field admissibility) but never defined. The downstream effect: the sectorwise zero-decomposition (Theorem 2.7.1) now uses M_{red} 's sectoral structure, giving the theorem genuine content (the weights p_k are constrained by M_{red} 's positivity and decomposition, not arbitrary). This is the load-bearing fix that strengthens the Born rule derivation (§2.7.4).

Definition 1.4.6 (Emergent Dirac Bracket — Algebraic)

The **Dirac bracket** on A_{red} is:

$$[A, B]_D := [A, B] - \sum_{a, b} [A, \hat{K}_a] \cdot (\Delta^{(1)})^{ab} \cdot [\hat{K}_b, B]$$

where $(\Delta^{(1)})^{ab}$ is the inverse of the Tier 1 compatibility matrix $\Delta^{(1)}_{ab} = [K_a, K_b]$. The Dirac bracket is the fixed-point effective commutator of the Open Extension descent (Theorem 1.6.1).

Theorem 1.4.7 (Dirac Bracket Properties)

The Dirac bracket $[\cdot, \cdot]_D$ on A_{red} satisfies:

1. **Antisymmetry:** $[A, B]_D = -[B, A]_D$.
2. **Jacobi identity:** $[[A, B]_D, C]_D + \text{cyclic} = 0$.
3. **Tier 1 vanishing:** $[A, K_a]_D = 0$ for all $A \in A_{\text{red}}$, $a \in I_1$ (the Tier 1 constraints are in the center of the Dirac bracket).
4. **Tier 2 preservation:** $[A, D_i]_D = [A, D_i]$ for all $A \in A_{\text{red}}$, $i \in I_2$ (the Tier 2 constraints retain their original bracket).

Proof sketch. (1) and (2) are standard properties of the Dirac bracket (Dirac 1964, Henneaux-Teitelboim 1992). (3) follows from the definition: $[A, K_a]_D = [A, K_a] - \sum_{b,c} [A, K_b] (\Delta^{(1)})^{bc} [K_c, K_a] = [A, K_a] - [A, K_a] = 0$ (using $(\Delta^{(1)})^{bc} [K_c, K_a] = \delta^b_a$). (4) follows because D_i has zero bracket with all Tier 1 constraints (by the Tier 1/Tier 2 split).

[GAP: v2.1 — the Dirac bracket's properties (Theorem 1.4.7) are standard, but their RBF-specific verification (that the Tier 1/Tier 2 split is clean, that $(\Delta^{(1)})^{ab}$ exists and is invertible, that the Jacobi identity holds on A_{red}) depends on the fracture being well-defined (GAP-1.2.4). If the fracture stalls (T-2 fails), the Dirac bracket reverts to a formal

construction without physical grounding. Flagged as GAP-1.4.7, severity MATERIAL.]

§1.5 The Reconciliation Propagator

Definition 1.5.1 (Reconciliation Propagator — Three-Mechanism Composition)

The **Reconciliation Propagator** is the dynamical engine that drives kinematic states toward the admissible sector:

$$R_t = \lim_{N \rightarrow \infty} \left(R_{T/N}^{\text{OExt}} \circ (R^{\text{OExp}})^N \right), \quad \text{quad}$$

where R^{OExp} is the Open Expansion flow (Definition 1.5.2) and R^{OExt} is the Open Extension flow (Definition 1.5.3). The propagator is the ordered composition of two scale-distinct mechanisms: Open Expansion (local, isometric) at the fast scale ε , and Open Extension (global, contractive) at the slow scale T/N .

Definition 1.5.2 (Open Expansion Flow — "was SOE")

The **Open Expansion flow** is the local, isometric expansion of locality-built sectors:

$$R^{\text{OExp}}_{\varepsilon}[\rho] = e^{i \sqrt{\hat{F}} + \delta \hat{F}} \varepsilon \rho, \quad e^{-i \sqrt{\hat{F}} + \delta \hat{F}}$$

where $\delta \hat{F} = \delta F(\delta \hat{C})$ is the incremental fracture update from the Open Expansion step. The generator is $H_{\text{eff}} = \sqrt{F + \delta F}$ — the Hamiltonian that drives the spectral-flux flow.

(was SOE: "Single Open Extension" in v1.0. Renamed "Open Expansion" in v2.0/v2.1 to capture the continuous, driven expansion of sectors toward reconciliation.)

Remark 1.5.3 (Open Expansion as Hamiltonian). The Open Expansion flow is Hamiltonian (unitary, isometric) — it preserves the spectrum of ρ . This is the "kinematic" component of the propagator: it expands the algebraic scope of sectors without dissipating burden. The generator $H_{\text{eff}} = \sqrt{F}$ has Lieb-Robinson velocity $v_{\text{LR}} = \ell_0 \cdot \sqrt{\lambda_{\min}(F)} = c$ (Theorem 1.1.4), confirming that Open Expansion respects the finite propagation speed.

Definition 1.5.4 (Open Extension Flow — "was MOE")

The **Open Extension flow** is the global, contractive extension of reconciliation across accumulated expansions:

$$R^{\text{OExt}}_t[\rho] = \rho - t \cdot \eta \cdot \nabla_{\text{Bures}} B_{\Delta}[\rho] + O(t^2)$$

where $\nabla_{\text{Bures}} B_{\Delta}$ is the Bures-gradient of the burden functional, and $\eta > 0$ is the reconciliation rate. At leading order (T-2 conditional):

$$\frac{d\rho}{dt} \Big|_{\text{OExt}} = -\eta \cdot [\hat{F}, [\hat{F}, \rho]]$$

which is Lindblad form with Lindblad operator $L = \sqrt{F}$.

(was MOE: "Multiple Open Extension" in v1.0. Renamed "Open Extension" in v2.0/v2.1 to capture the global, contractive extension of reconciliation across the accumulated expansions.)

Remark 1.5.5 (Open Extension as Lindblad dissipator). The Open Extension flow is dissipative (Lindblad) — it reduces burden by gradient descent on the Bures metric. This is the "dynamic" component

of the propagator: it reconciles the sectors by driving the global state toward the Master-Zero fixed point. The Bures metric is the natural metric for this descent (Petz 1996 classification of monotone quantum metrics — see §2.7 for the connection to the Born rule).

[GAP: v2.1 — the Lindblad form $d\rho/dt = -\eta[F, [F, \rho]]$ is T-2-conditional (flagged in v2.0 as Theorem Target T-2). Without T-2, the dissipative structure of R_t is not rigorously established. This dependency propagates to: the Arrow of Time proof (§4.1.4, uses Spohn's inequality on the Lindblad generator) and the decoherence mechanism (§2.6.3, uses cross-sector burden from Open Extension). The v2.1 patch plan (M8) recommends adding inline [T-2] tags to these downstream theorems. Flagged as GAP-1.5.4, severity MATERIAL (propagates).]

Definition 1.5.6 (Dephasing — Residual Cross-Eigenspace Suppression)

The **dephasing** mechanism suppresses residual cross-eigenspace coherences that remain after Open Expansion and Open Extension:

$$R_{\text{varepsilon}}^{\text{Deph}}[\rho] = \sum_k P_k \rho P_k + P_{\text{cross}} \rho P_{\text{cross}}$$

This projects ρ onto the block-diagonal form (in the sector decomposition), eliminating off-diagonal coherences between sectors that have decohered. Dephasing is the third mechanism in the propagator composition, acting at the residual scale.

Remark 1.5.7 (Terminology rename summary)

v1.0 term	v2.0/v2.1 term	Reason
---	---	---
SOE (Single Open Extension)	Open Expansion	Captures continuous, driven expansion of locality-built sectors
MOE (Multiple Open Extension)	Open Extension	Captures global, contractive extension of reconciliation across expansions

The mathematics is unchanged; the physical interpretation is sharpened. The "(was SOE)" / "(was MOE)" parentheticals in the definition headers provide traceability.

§1.6 The Convergence Theorem

Theorem 1.6.1 (Convergence to the Physical Sector)

[v2.1: EXPANDED proof strategy relative to v2.0. v2.0 stated the theorem with a LaSalle invariance proof sketch verified only in a finite-dim toy model ($M_6(\mathbb{C})$). v2.1 expands the strategy: F is a frustration-free Hamiltonian; the detectability lemma (Aharonov et al. 2009) lifts local gaps to a global gap.]

The Reconciliation Propagator R_t drives every kinematic state to the physical sector:

$$\lim_{t \rightarrow \infty} \rho_t = \frac{P_0 \rho_0 P_0}{\text{Tr}(P_0 \rho_0)}$$

where P_0 projects onto $\ker(M) = \ker(F)$, and $\rho_t = R_t[\rho_0]$.

Status: Conditional (T-2 for the infinite-dimensional case). The finite-dimensional case is verified ($M_6(\mathbb{C})$ toy model: $\text{Tr}(\rho_0 M) = 0.96 \rightarrow 2.14 \times 10^{-4}$).

Proof strategy (v2.1 expanded).

Step 1: Frustration-free structure. $F = \sum_{\alpha,\beta} f_{\alpha\beta}$ where $f_{\alpha\beta} = \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta}) \geq 0$. The ground state space is $\ker(F) = \bigcap_{\alpha,\beta} \ker(f_{\alpha\beta}) = \ker(M)$. This is the standard frustration-free Hamiltonian structure (Nachtergaele 1996).

Step 2: Local finite-dimensionality. Each local sector H_k is finite-dimensional (bounded by the ℓ_0 -floor: each sector has $\leq N_{\text{cell}} \sim \text{vol}/\ell_0^3$ reconciliation cells). On finite-dim H_k , the local gap $\gamma_{\text{local}} > 0$ is guaranteed by LaSalle (standard Lyapunov theory for gradient descent on a Morse function — the burden functional B_Δ is Morse on H_k).

Step 3: Detectability lemma (Aharonov-Arad-Landau-Vazirani 2009). If the local gap γ_{local} is uniform across sectors, and the constraint adjacency graph G_I (Definition 1.1.6) has bounded degree Δ , then the global spectral gap satisfies:

$$\gamma_{\text{global}} \geq \frac{\gamma_{\text{local}}}{C \cdot \Delta}$$

where C depends on the commutation structure of the local terms. Since $D=3$ is fixed (§3.6, conditional), $\Delta \leq 6$ (each cell has at most 6 neighbors in 3D), so:

$$\gamma_{\text{global}} \geq \frac{\gamma_{\text{local}}}{6C} > 0$$

Step 4: Cross-sector decay. The Lieb-Robinson bound (Theorem 1.1.4) gives exponential decay of correlations outside the causal cone: $|\langle \psi | [A_x, B_y] | \psi \rangle| \leq C \|A\| \|B\| e^{-\mu d(x,y)}$ with $\mu \propto \gamma_{\text{global}}$. This ensures the thermodynamic limit ($N \rightarrow \infty$ sectors) preserves convergence — the cross-sector influence decays exponentially with distance.

Step 5: Global convergence. Combining Steps 2–4: the local gap (Step 2) lifts to a global gap (Step 3), and the cross-sector decay (Step 4) ensures the global gap doesn't close in the thermodynamic limit. The Convergence Theorem follows: R_t drives every state to $\ker(M) = \ker(F)$.

Remark 1.6.2 (What T-2 actually requires). T-2 is NOT the demand for a spectral gap on an abstract infinite-dimensional H_{kin} . It is the demand for:

1. **Local gap:** $\gamma_{\text{local}} > 0$ on each finite-dimensional sector H_k . (Plausible — needs verification on the specific constraint algebra, not just the $M_6(\mathbb{C})$ toy model.)
2. **Uniformity:** γ_{local} doesn't shrink as new sectors are added. (Plausible because the local algebra per cell is always $M_d(\mathbb{C})$ — but needs proof.)
3. **Bounded degree:** the adjacency graph G_I has bounded degree Δ . (GAP-1.1.7 — FATAL, blocks T-2.)

If all three hold, the detectability lemma gives the global gap, and the Convergence Theorem is Established. If any one fails, T-2 remains open.

[GAP: v2.1 — the three prerequisites (local gap, uniformity, bounded degree) are the open components of T-2. The bounded degree (GAP-1.1.7) is the most critical — without it, the detectability lemma doesn't apply. The uniformity condition is the key remaining technical step even if bounded degree is established. Flagged as GAP-1.6.1, severity FATAL (T-2 is the existential open problem).]

Corollary 1.6.2 (Master-Zero Equivalence — "DERIVED, not postulated")

The Master-Zero condition $\omega(M) = 0$ is a derived asymptotic property, not a postulate:

$$\ker(\hat{M}) = \ker(\hat{F})$$

Proof. The forward inclusion $\ker(M) \subseteq \ker(F)$ is algebraic (Remark 0.3.6: if $\omega(M) = 0$ then all $\hat{C}_\alpha$ vanish, so all $\Delta_{\{\alpha\beta\}}$ vanish, so $F = 0$). The reverse inclusion $\ker(F) \subseteq \ker(M)$ follows from the Convergence Theorem (Theorem 1.6.1): R_t drives every state to $\ker(F)$, and the fixed point is $\ker(M)$.

Corollary 1.6.3 ($\Lambda_B = 0$)

The bare cosmological constant vanishes: $\Lambda_B = 0$.

Proof. $\Lambda_B = 0$ because the Master-Zero condition $\omega(M) = 0$ is an exact asymptotic attractor (Corollary 1.6.2), not a tuned parameter. There is no vacuum energy to fine-tune.

Status: Conditional on T-2 (inherits from Theorem 1.6.1). The v1.0 "EXACT" branding is reverted to "[Conditional on T-2]."

Theorem 1.6.4 (Local Convergence, Global Desynchronization)

[NEW in v2.1 (was informally stated in v2.0). This theorem gives the local-convergence / global-desynchronization structure its first formal statement.]

The asymptotic state is a desynchronized composite:

$$\rho_\infty = \bigoplus_k p_k \cdot \rho_\infty^{\{(k)\}} + \rho_{\text{cross}}$$

where each $\rho_\infty^{\{(k)\}} \in \ker(M) \cap H_k$ is a locally converged sector, and ρ_{cross} is the residual cross-sector coherence. The residual cross-sector burden:

$$B_{\text{cross}}[\rho_\infty] = \text{Tr}(\rho_\infty \cdot \hat{F}_{\text{cross}})$$

is the quantity that, at cosmological scale (§6), appears as dark energy — the cost of maintaining the globally desynchronized composite.

Proof sketch. By the Convergence Theorem (Theorem 1.6.1), each sector H_k converges locally to $\ker(M) \cap H_k$. But the finite propagation speed $c = \gamma \cdot \ell_0$ (Theorem 1.1.4) prevents global synchronization — different sectors converge to potentially different local fixed points. The global state is therefore a direct sum (with weights p_k) of locally converged sectors, plus residual cross-sector coherence. The cross-sector burden B_{cross} is non-zero because the sectors are desynchronized.

Remark 1.6.5 (Dark energy as global desynchronization cost). Theorem 1.6.4 is the algebraic seed of the dark-energy mechanism (§2.2, §6.3). Dark energy is not a cosmological constant ($\Lambda_B = 0$, Corollary 1.6.3) — it is the residual burden of maintaining the globally desynchronized composite. The quantitative form $\rho_{\text{DE}} \sim (H/\Gamma_{\text{eff}})^2$ is developed in §2.2; the cosmological application is in §6.3.

§1.7 The Thin/Full Physical Sub-Algebra Split

Definition 1.7.1 (Thin Physical Sub-Algebra)

The **thin physical sub-algebra** is:

$$A_{\text{phy}}^{\text{thin}} := \{A \in \mathcal{A} : \pi_{\text{kin}}(A) \in \ker(\hat{M})\} \subseteq \ker(\hat{M})$$

This is the set of observables that preserve the physical kernel — the "thin candidate" for the physical sub-algebra, defined using only $\ker(M)$ (available after the Convergence Theorem, §1.6).

Definition 1.7.2 (Thin Candidate Properties)

The thin sub-algebra $A_{\text{phy}}^{\text{thin}}$ is:

1. A $*$ -sub-algebra of A (closed under product and involution).
2. Contains the identity ($1 \in A_{\text{phy}}^{\text{thin}}$).
3. Acts on $\ker(M)$ by left-multiplication.

Definition 1.7.3 (Full Physical Sub-Algebra)

The **full physical sub-algebra** is:

$$A_{\text{phy}}^{\text{full}} := \{A \in A : \pi_{\text{kin}}(A) \setminus \ker(\hat{M}_{\text{red}}) \subseteq \ker(\hat{M}_{\text{red}})\}$$

defined using $\ker(M_{\text{red}})$ (the reduced master constraint, Definition 1.4.3) on A_{red} .

Theorem 1.7.4 (Thin = Full)

The thin and full physical sub-algebras are equal:

$$A_{\text{phy}}^{\text{thin}} = A_{\text{phy}}^{\text{full}}$$

Status: Conditional on T-2 (the equality requires the Convergence Theorem to establish that $\ker(M) = \ker(M_{\text{red}})$ on A_{red}).

Proof sketch. The thin candidate uses $\ker(M)$ (the full master constraint); the full candidate uses $\ker(M_{\text{red}})$ (the reduced master constraint on A_{red}). By the Tier 1/Tier 2 split (§1.4), $M = M_{\text{Tier1}} + M_{\text{red}}$, and the Tier 1 part is quotiented away on A_{red} . The Convergence Theorem (Theorem 1.6.1) establishes that both kernels coincide on A_{red} (the fixed point is the same), so the two candidates are equal.

Remark 1.7.5 (Why this matters). Theorem 1.7.4 is referenced by:

- The field admissibility condition (§2.3.1): fields preserve $\ker(M_{\text{red}})$.
- The particle definition (§2.4.2): particles are stable modes in $A_{\text{phy}}^{\text{full}}$.
- The mass-burden identity (§2.5.1): $m^2 \equiv B_0$ uses the ground-state maintenance burden in A_{phy} .

The thin/full equality ensures these definitions are consistent — there is one physical sub-algebra, not two competing candidates.

§1.8 Zero-Preserving Events and the Causal Order**Definition 1.8.1 (Zero-Preserving Event)**

A **zero-preserving event** is an element $e \in A_{\text{phy}}^{\text{thin}}$ such that:

$$\omega_{\text{kin}}(e^{\dagger} e) > 0 \quad \text{and} \quad \omega_{\text{kin}}(e^{\dagger} \hat{M} e) = 0$$

The first condition ensures e has non-vanishing GNS norm (it "exists"); the second ensures e preserves the Master-Zero condition (it is "physical"). The set of zero-preserving events is denoted E_{phy} .

Remark 1.8.2 (Events are defined after thin/full, not before). In v1.0, events were defined at §1.1.1 — before the thin/full split existed. This was internally inconsistent because the event definition requires $A_{\text{phy}}^{\text{thin}}$. v2.0/v2.1 corrects this: events are defined at §1.8.1, after the thin/full split (§1.7). The relocation is forced by acyclicity.

Definition 1.8.3 (Reconciliation Depth)

The **reconciliation depth** of an event e is:

$$d(e) := \text{the number of Open Expansion steps required to incorporate } e \text{ into the}$$

This is a simple step-count (v2.0/v2.1 simplification: v1.0's two-scale formula $d(E) = d_{\text{SOE}} + N \cdot d_{\text{MOE}}$ is dropped because the SOE/MOE distinction is now terminological, not structural).

Definition 1.8.4 (Primitive Causal Relation)

The **primitive causal relation** $<$ is defined on E_{phy} :

$$e_1 \prec e_2 \quad \Longleftrightarrow \quad d(e_1) < d(e_2) \quad \text{and} \quad e_2 \text{ depends on } e_1$$

where "depends on" means that e_2 's admissibility requires e_1 to have been incorporated first (e_2 cannot be a zero-preserving event without e_1 preceding it).

Assumption 1.8.5 (Causal Partial Order)

The relation $<$ is a **strict partial order** on E_{phy} :

1. **Irreflexive:** $e \not< e$ for all e .
2. **Antisymmetric:** if $e_1 < e_2$ then $e_2 \not< e_1$.
3. **Transitive:** if $e_1 < e_2$ and $e_2 < e_3$ then $e_1 < e_3$.

Status: Structural assumption (the strict partial order properties are algebraically motivated but not fully proven from the Lie algebra structure — this is T-1/T-2 dependent).

Remark 1.8.6 (Causal order is operational, not temporal). The causal order $<$ is an operational ordering on zero-preserving events — it captures dependency, not temporal sequence. Duration (the operational "when") is constructed in §4.1 from burden-weighted causal depth. The "now" (the joint where + when) is defined in §4.2. Here in §1.8, we have only the structural ordering $<$, which is the Lie primitive (§0.4.1) made operational.

§1.9 Architectural Summary

What Section 1 Contains

Subsection	Content	Layer
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§1.0	Purpose: the dynamical foundation	—

Subsection	Content	Layer
§1.1	Open Expansion Principle, Lie–Jordan mismatch, finite rate, adjacency structure	L→C
§1.2	Fracture of H_{kin} , sector projectors, F decomposition	C
§1.3	Burden, burden linearity, B_{intra} + B_{rel} decomposition	C
§1.4	Tier 1/Tier 2 split, A_{red} , M_{red} , Dirac bracket	C
§1.5	Reconciliation Propagator (Open Expansion ◦ Open Extension ◦ Dephasing)	C
§1.6	Convergence Theorem, Master-Zero equivalence, $\Lambda_B = 0$, global desynchronization	C
§1.7	Thin/full A_{phy} split	C
§1.8	Zero-preserving events, reconciliation depth, causal order $<$	C

What Section 1 Does NOT Contain (Deferred)

Object	Deferred to	Reason
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Reconciliation Principle	§2.1	Requires burden (§1.3) and events (§1.8) — both now available
Correlation kernel K_{ω} / K_{ω}^J	§3.1	Requires A_{loc} (§3.1, which requires $A_{\text{phy}}^{\text{thin}}$ from §1.7)
Fubini-Study metric	§3.2	Requires K_{ω} (§3.1)
Π_{net} (cross-extension network operator)	§3.9	Requires K_{ω} (§3.1) and sectoral decomposition (§1.2)
Duration, burden-clock $\alpha(B)$	§4.1	Requires the "now" (§4.2) which requires space (§3) + time (§4.1)
Burden tensor $\Theta^{(B)}_{\mu\nu}$	§4.3	Requires the emergent metric (§3)
Einstein closure, gravity	§4.5	Requires burden tensor (§4.3)
Three-tier burden decomposition (mode + relational + interaction)	§4.4	Requires F decomposition (§1.2.4) — now available
Dark energy mechanism	§2.2	Requires burden (§1.3) and global desynchronization (§1.6.4) — both now available

Acyclicity Test

Question: Does Section 1 define every object before using it?

Verification:

- §1.1 uses A, M, F, A_L, A_J (from §0). ✓
- §1.2 uses F, ω_{kin} (from §0, §1.1). ✓
- §1.3 uses F, F_intra, F_cross (from §0, §1.2). ✓
- §1.4 uses F, M, $\Delta_{\alpha\beta}$ (from §0, §1.2). ✓
- §1.5 uses F, B_Δ, H_kin (from §0, §1.3). ✓
- §1.6 uses R_t, M, F (from §1.5, §0). ✓
- §1.7 uses M, M_red, A_red (from §0, §1.4). ✓
- §1.8 uses A_phy^{thin}, M (from §1.7, §0). ✓

Forward references:

- §1.2.6 references Π_{net} (§3.9) — named as a forward pointer, not used in any definition.
- §1.5.5 references the Bures metric and Petz classification (§2.7) — named as context, not used.
- §1.6.4 references dark energy (§2.2, §6.3) — named as the downstream consequence, not used.
- §1.8 references duration (§4.1) and the "now" (§4.2) — named as deferred, not used.

No forward reference creates a circularity.

Verdict: Section 1 is acyclic. The architecture holds through the dynamical foundation.

Inline Gap Summary for §1

Gap ID	Location	Description	Severity	Blocks
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GAP-1.1.3	§1.1 Remark 1.1.3	δJ formal decomposition (J_target vs J_current) not rigorous	LOW	Cosmetic ($\delta J = -B_{\Delta}$ is consistent)
GAP-1.1.7	§1.1 Remark 1.1.7	Adjacency graph G_I bounded degree not proven	FATAL	T-1 (§1.1.4), T-2 (§1.6.1)
GAP-1.2.4	§1.2 Remark 1.2.6	Fracture existence depends on T-2	MATERIAL	F decomposition (§1.2.4), relational burden (§4.4)
GAP-1.4.7	§1.4 Remark 1.4.7	Dirac bracket RCF-specific verification depends on fracture	MATERIAL	Dirac bracket (§1.4.6)

Gap ID	Location	Description	Severity	Blocks
GAP-1.5.4	§1.5 Remark 1.5.5	Lindblad form T-2-conditional; not propagated to downstream	MATERIAL	Arrow of Time (§4.1.4), decoherence (§2.6.3)
GAP-1.6.1	§1.6 Remark 1.6.2	T-2 three prerequisites (local gap, uniformity, bounded degree) open	FATAL	Convergence Theorem (§1.6.1), all T-2-conditional results

Total new gaps surfaced in §1: 6 (2 fatal, 3 material, 1 low). Of these:

- GAP-1.1.7 and GAP-1.6.1 are the same underlying issue (bounded degree + T-2 prerequisites) — the existential open problem.
- GAP-1.2.4, GAP-1.4.7, GAP-1.5.4 are all T-2-propagated dependencies — they resolve when T-2 closes.
- GAP-1.1.3 is a notational cleanup ($\delta J = -B_\Delta$ is consistent but not elegant).

No NEW structural gaps introduced by the v2.1 rewrite of §1. All 6 gaps are pre-existing from the Phase C program, now flagged at their point of origin.

Section 1 is CLOSED. Section 2 (Burden, Matter, and the Reconciliation Principle) can be merged against this stable dynamical foundation.